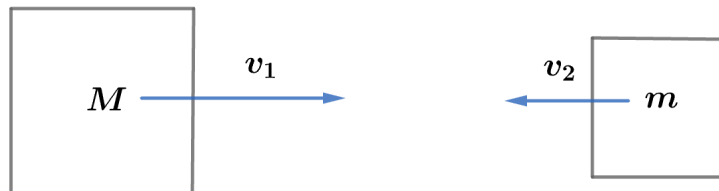


2012 F=ma Exam: Problem 5

Kevin S. Huang



The initial kinetic energy is given by

$$K_i = \frac{1}{2}Mv_1^2 + \frac{1}{2}mv_2^2 = \frac{1}{2}(12\text{ kg})(4\text{ m/s})^2 + \frac{1}{2}(6\text{ kg})(2\text{ m/s})^2 = 108\text{ J}$$

Conserving linear momentum, we have

$$Mv_1 - mv_2 = (M + m)v_f$$
$$v_f = \frac{Mv_1 - mv_2}{M + m} = \frac{(12\text{ kg})(4\text{ m/s}) - (6\text{ kg})(2\text{ m/s})}{12\text{ kg} + 6\text{ kg}} = 2\text{ m/s}$$

where v_f is the final velocity of both masses. The final kinetic energy is given by

$$K_f = \frac{1}{2}(M + m)v_f^2 = \frac{1}{2}(18\text{ kg})(2\text{ m/s})^2 = 36\text{ J}$$

The loss in kinetic energy is

$$|\Delta K| = K_i - K_f = 72\text{ J}$$

so the answer is D.